

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

May-2014

EC 209 Microprocessors & Microcontroller

Date: 03.05.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

Q - 1 Do as directed.

- (i) Calculate the number of memory chips needed to design 8K- byte memory if [02]  
the available memory chip size is 1024 x 1.
- (ii) How many address lines are necessary on the chip of 1MB memory? [02]
- (iii) Explain the function of ALE pin in 8085. [01]
- (iv) The memory address of the last location of an 8K byte memory chip is FFFFh, [02]  
Find the starting address.

- Q - 2 (a) Interface 4KB of RAM with address range B000h to BFFFh and 4KB of ROM with [07]  
address range F000h to FFFFh. (Use only one 3 to 8 Decoder).
- (b) Draw the timing diagram of IN 84h. [07]

## OR

- Q - 2 (a) In figure (1), specify the memory range of ROM 1, ROM 2 and RAM 1. [07]
- (b) Differentiate between memory mapped I/O and peripheral mapped I/O. [07]
- Q - 3 (a) Draw the block diagram of 8255 and explain control word and mode 1 with suitable [07]  
diagram.
- (b) Interface one output port with address F8h and one input port with address FAh. [07]

## OR

- Q - 3 (a) Explain flag register of 8051 with function of each flag. [04]
- (b) Explain internal RAM organization of 8051. [03]
- (c) Explain counters and timers of 8051. Explain timer mode 1 and mode 2. [07]

## SECTION – II

Q - 4 Do as directed

(i) Identify the addressing mode of following instruction. [02]

(1) Mov r0, #0ACh

(2) Push 30h

(ii) Copy the external code byte at address 007Dh to the SP. [02]

(iii) Complement the content of accumulator without using CPL instruction. [01]

(iv) Clear bit 3 of RAM location 22h without affecting any other bit. [01]

(v) What will be the content of SP after executing following instructions? [01]

MOV SP, #19h

PUSH 07H

PUSH 23H

Q - 5 (a) Write a program to put a random number in R3 and increment it until it equals E1h. [03]

(b) Write a program to convert any HEX number to BCD. [05]

(c) Write a program to sort block of data in ascending order. [06]

OR

Q - 5 (a) Write a program to generate 2KHz square wave on P1.1. (Use timer delay and crystal frequency is 11.0592MHz). [07]

(b) Write a program to transmit "Microprocessors and Microcontroller" serially at baud rate of 9600. (SMOD=1 and crystal frequency is 11.0592MHz). [07]

Q - 6 (a) Explain calls and subroutines with sequence of events. [07]

(b) Write a program to receive data serially with baud rate 4800 and send received data on P1. (Use crystal frequency = 11.0592MHz). [07]

OR

Q - 6 (a) Write a program to generate 100Hz square wave on P1.1 and 200Hz square wave on P1.2 using timer interrupt method. (Use crystal frequency = 12MHz). [07]

(b) Write a program to Count number of interrupts arriving on external interrupt pin INT1. Stop when counter overflows and disable the interrupt. [07]

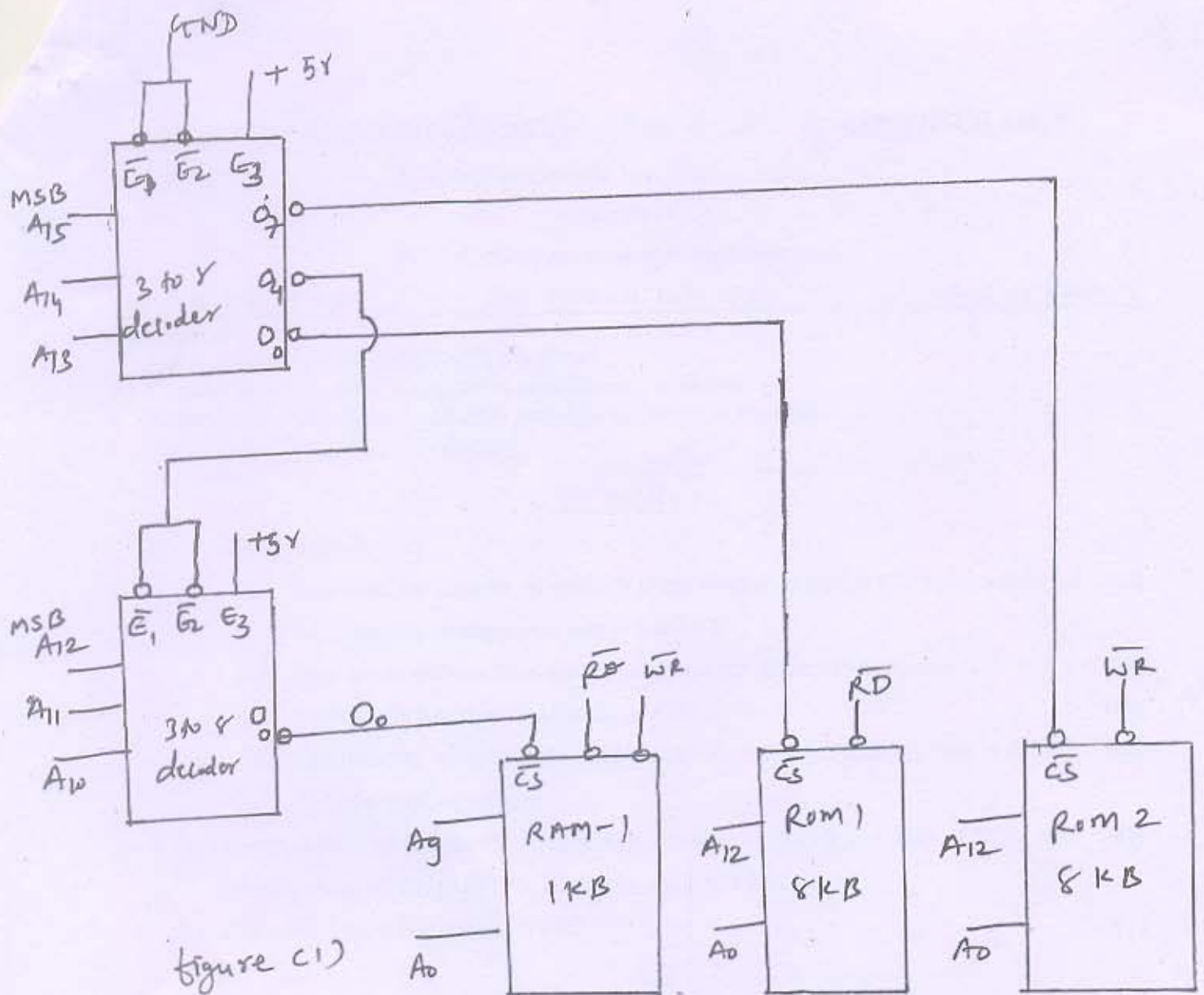


figure c1)